

The 10th International Conference on Computer Science and Computational Intelligence 2025

Driving User Acceptance of AI-Enhanced Weather Applications: A Behavioral Study Using UTAUT2 and Trust

Marvin Luckianto Nahtan^{a*}, Stenly Lukmana^a, Azani Cempaka Sari^a, Andien Dwi
Novika^a, Immanuel Lucky Pratama Sugiharto Rusli^a

^aComputer Science Department, School of Computer Science, Bina Nusantara University, Jakarta, Indonesia 11530

Abstract

Weather applications are widely used to support daily activities, yet users continue to face challenges related to trust, personalization, and overall engagement. Artificial intelligence (AI) offers opportunities to improve both usability and forecast accuracy in weather apps, particularly through features such as chatbots, contextual recommendations, and personalized weather information. Despite these advancements, there is limited empirical understanding of how AI-based features influence user acceptance compared to traditional weather applications. This study evaluates user acceptance of a weather app prototype enhanced with AI capabilities, using the Unified Theory of Acceptance and Use of Technology 2 (UTAUT2) as the primary framework, extended by the Trust construct. A quantitative and qualitative approach was employed through a questionnaire that aims to identify whether AI features can enhance user satisfaction, and which features are most valuable for future development of weather applications. A mixed-method approach was employed involving a comparative questionnaire completed by 101 participants, assessing both a generic weather app and the AI-enhanced prototype. The results reveal that the AI-based prototype significantly outperforms generic apps in several UTAUT2 constructs, including Performance Expectancy, Trust, Hedonic Motivation, Habit, and Social Influence. Qualitative feedback emphasizes the value of AI-driven personalization (e.g., clothing suggestions, activity planning), real-time accuracy, and calendar integration. These findings suggest that AI capabilities can meaningfully improve user satisfaction, trust, and behavioral intention to use weather apps. The study provides key insights for future design and development of intelligent, user-centered weather applications.

© 2025 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<https://creativecommons.org/licenses/by-nc-nd/4.0>)

Peer-review under responsibility of the scientific committee of the 10th International Conference on Computer Science and Computational Intelligence (ICCCSI) 2025

Keywords: Artificial intelligence; Weather applications; User acceptance; UTAUT2; User experience; User engagement

* Corresponding author.

E-mail address: marvin.nahtan@binus.ac.id

1. Introduction

Weather is one of the biggest factors in daily human lives, from planning outdoor activities to managing health and well-being. Although weather apps are widely used, studies show that users often experience issues with weather apps related to trust, perceived accuracy, and engagement [1]. Many people rely on weather apps for forecasts, but usually only interact with basic features, such as daily and hourly predictions, while more advanced and personalized features are used less, resulting in information that lacks context and personalization [2]. This lack of personalization and interactivity remains a challenge to improving user acceptance.

With the growing presence of Artificial Intelligence (AI) in technology, there's a new opportunity to make weather apps more useful and user centered. AI chatbots have changed the communication landscape and are already helping users by enhancing access to information [3]. AI systems adjust output and responses according to users' preference and personalization, significantly improving engagement and user satisfaction [4-6]. Moreover, explainability plays a big role in user trust and satisfaction. Users are more likely to rely on a system that can make them understand why a prediction, recommendation, or answer was made [6]. These capabilities are relevant for weather applications, where context is important, rather than just raw data, which helps in making decisions such as clothing recommendations, event planning, safety alerts, and many more personalized actions.

In addition to improving usability, AI also has the potential to increase forecast accuracy. Machine learning models can process large amounts of historical and real-time data to predict weather more reliably than traditional methods such as the Pangu-Weather Model [7-8]. Studies proved that AI weather models are more robust under changing climate conditions and offer speed and accuracy which outperform traditional weather prediction [9-11]. As a result, many researchers have begun focusing on AI-based forecasting to make weather information more precise and relevant.

Despite AI advancements in weather apps, there are relatively few studies addressing how AI based features can influence user acceptance of weather apps compared to generic apps. Existing research primarily focuses on usage of AI and AI chatbots in services and applications to increase user experience [3-6, 10, 13], studying weather app usage and determining the factors that affect user acceptance [1-2, 15, 16] or improving technical forecast [7-11] without bridging them. Given this gap, this study aims to evaluate whether integrating AI into a weather application prototype can improve user acceptance compared to generic weather apps. The comparison will focus on aspects such as usability and overall user satisfaction. Furthermore, this research seeks to understand which features matter most to users, offering insights that could guide future improvements in weather app design and functionality.

2. Literature Review

2.1. Weather Applications

Weather applications are widely used everyday by people all over the world for general forecast information, but there is still a gap in effective engagement and interpretation. The majority of users interpret forecasts on their own, making them ambiguous and leading to false expectations [2]. Low trust and value are still relevant despite high usage and availability of weather applications [1]. To address this, weather services are integrating digital tools to increase value and user engagement, but user engagement is often inconsistently defined by these services, some think it just means giving users more information, others think it means letting users be involved in shaping the service. This leads to engagement strategies that fail to meet user needs [16].

2.2. User Acceptance Models

For years, researchers have relied on framework models to see how users would engage with systems and technologies to better explain and evaluate user acceptance. TAM is one such model, taking measurements in Perceived Usefulness (PU), Perceived Ease of Use (PEU), Attitude to Use (AU), and Intention to Use (IU) [18]. TAM is foundational in measuring user acceptance for traditional applications. However, it is insufficient for more modern applications that have started integrating AI. A newer model, UTAUT2, is a more efficient framework to measure user acceptance in more modern applications. UTAUT2 takes measurements in Performance Expectancy (PE), Effort Expectancy (EE), Social Influence (SI), Facilitating Conditions (FC), Hedonic Motivation (HM), Price Value (PV),

and Habit (H) [17]. UTAUT2 is able to evaluate complex motivation and contextual influences that shape how users interact with AI-integrated applications. Aspects such as PE, SI, and H are very strong in predicting user acceptance [19], while Zhang et al. (2022) [17] concluded that UTAUT2 is a suitable framework in measuring user acceptance for service based applications. Yildiz Durak & Onan (2024) [5] also showed the effectiveness of UTAUT2 in chatbots, in which PE and SI are essential.

2.3. UTAUT2

Trust from the user base is a critical component in their willingness to use AI-based applications. Trust in AI is shaped by accuracy, transparency, and fairness. Users need to believe that the system will perform as expected since AI systems introduce uncertainty. The lack of transparency in AI systems contributes to the users base's difficulty in trusting AI based applications [12]. Meanwhile AI systems which incorporate user-centered design features are more likely to gain trust from the user base [6]. Trust and privacy concerns are significant predictors of the user base's behavioral intentions [13], as trust is one of the strongest predictors of behavioral intention when using AI based systems [5]. Trust from the user base is indispensable, if users see an inconsistency or inaccuracy in weather forecasts, they may or may not abandon the application and/or distrust its functionalities [21], as trust plays a critical role in the user base's willingness to engage in AI-integrated technologies [14]. Forecast contradictions between weather forecasts can and will erode user trust [15].

2.4. UX and Design as Drivers of Acceptance

In order to design an AI-integrated weather application that is not only accurate but also interactive, it's crucial to have an understanding of the factors influencing the user experience (UX). Factors such as output quality, perspicuity, dependability, and novelty shape the core acceptance drivers such as EE, HM, and behavioral intention. EE, relates to how easy the user base believes a system is to use. Ease of navigation, clarity of interface, and customization options enable users to quickly find relevant forecasts or alerts, reinforcing the user base's perception of usability and efficiency [1]. Ease of use aside, hedonic motivation, how enjoyable the interaction feels, also plays a pivotal role in the UX. Perspicuity, novelty and dependability all improve user acceptance by shaping the user base's emotional response [20]. Applications that are more visually engaging are able to build a more enjoyable user experience, which in result increases the user base's willingness to keep using said application, as interactivity and interface design are important to user acceptance of AI-tools [3]. Poor interface design can directly affect the perceived accuracy of even the most advanced AI-tools, many users misinterpret information from forecasts due to poorly designed communication and interface [2]. Personalized features allow for seamless trust building and a feel for easy accessibility.

2.5. AI Weather Forecasting

Recent breakthroughs in AI research has paved the way for more advanced AI-integrated applications. Advances in deep learning models, especially Convolutional Neural Networks (CNNs), can be applied effectively for weather prediction algorithms. Applying CNNs on a cubed sphere grid can improve global weather forecasting, which is a marginal improvement from the more traditional physics-based numerical weather prediction (NWP) models [10]. Leading AI models such as Pangu Weather, GraphCast, and AIFS are the best performing models when it comes to weather forecasting. Pangu Weather delivers high-resolution 3D forecasts at great speed and accuracy [7]. Meanwhile GraphCast and AIFS are advancing the field by delivering high-fidelity forecasts in minutes, being able to outperform leading NWP models [11]. These models are not only more efficient than traditional NWP models, they also provide robustness across multiple scenarios [9], showing infinite potential in serving as the next generation of weather forecasting. However, a challenge in implementing these models lies in their accessibility and trustworthiness. These AI-driven models must provide accurate outputs that are easily understandable [8]. Having inconsistencies across forecast applications damage the user base's trust in these applications [15]. In order to provide a much smoother integration into society, these models must provide improvements in their precision and consistency while being accompanied by user-centered design as well as communication strategies [21].

3. Methodology

This research uses a descriptive and correlational design to get user perception of AI based and generic weather apps as well as to examine the relationship between UTAUT2 variables and users' intention to use the weather app. This research will be conducted in three separate steps: Building prototype and questionnaire, handing out questionnaires, and evaluating as well as analyzing data.

3.1. Steps Overview

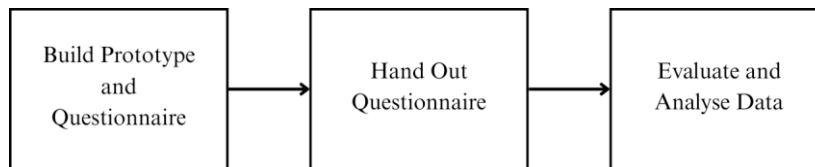


Fig. 1. Overview of our three-step research.

The research will be conducted in three main stages: building the prototype and questionnaire, handing out the questionnaire, as well as evaluating and analyzing the data. These steps are visually summarized in Figure 1, which represents an overview of the process.

1. **Build Prototype and Questionnaire:** The prototype is designed in figma and is designed as any other weather app, but it includes features such as AI-based personalization that can help with clothing recommendation, best places for weather and personalized local weather data including its context, via an AI chatbot. While the prototype does not implement real AI functionality, the design simulates AI-enhanced behavior through the UI. Calendar-aware suggestions are also presented visually to represent how an AI can help with suggestions based on a specific date's predicted weather and listed activities on said date. These features were clearly described to participants prior to evaluation to ensure accurate understanding of their intended functionality. The questionnaire is designed to collect quantitative and qualitative data. Quantitative data includes data related to UTAUT2 variables using a 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree), while qualitative data includes feedback from users containing features that users find helpful/important and recommendations for future development. The questionnaire is divided into 3 sections, the first is used to gather general data (age, employment status, etc.), the second is used to evaluate current existing generic weather apps, while the last is used to evaluate the prototype we built.
2. **Hand Out Questionnaire:** The questionnaires are distributed online through Google Forms. The questionnaire participants will mainly be university students with basic mobile app and technology literacy.
3. **Evaluate and Analyze Data:** Data received from the questionnaire are structured and unstructured data. Structured data are processed with descriptive statistics, while unstructured data are processed through thematic analysis.

3.2. Framework Overview

To evaluate user acceptance of AI-based weather applications, this study adopts the Unified Theory of Acceptance and Use of Technology 2 (UTAUT2) as the guiding framework. UTAUT2, proposed by Venkatesh, Thong, and Xu in 2012, extends the original UTAUT2 model by incorporating additional constructs: hedonic motivation, price value, and habit tailored to better understand user behavior from a consumer technology perspective [17]. Additionally, this study incorporates Trust as an external variable, as seen in illustrated in Figure 2, which is a critical factor in influencing users' confidence in the reliability, accuracy, and transparency of the app's predictions and recommendations [19].

1. **Performance Expectancy (PE):** The degree to which a user believes that using a system will help them in a task. It is one of the most significant predictors of a user's intention to use a system or technology. Performance Expectancy directly affects Behavioral Intention [17, 19].

2. **Effort Expectancy (EE):** Describes the degree of ease associated with the use of the system. If a system is perceived as easy to learn or operate, users are more likely to use it. *Effort Expectancy* directly *affects* Behavioral Intention [17, 19].
3. **Social Influence (SI):** The extent to which individuals perceive that people around them (family, friends, etc) believe they should use a particular system or technology. Social norms and opinions can significantly influence behavioral intention [17, 19].
4. **Facilitating Conditions (FC):** Refers to the user's availability of resources and support (device, internet access, knowledge, etc) to use a particular system or technology. Facilitating Conditions directly *affects* Use Behavior [17, 19].
5. **Hedonic Motivation (HM):** Refer to the level of enjoyment and pleasure from using the system or technology. This construct plays a strong role in consumer and mobile technology, where user experience is key. Hedonic Motivation directly *affects* Behavioral Intention [17, 19].
6. **Price Value (PV):** The user's evaluation of the drawbacks between the benefits (it being a monetary or non-monetary costs) from using the system or technology. It influences behavioral intention mostly when cost is a relevant factor [17, 19].
7. **Habit (HT):** The extent to which a user tends to perform a behavior automatically because of learning or repetition. A strong habit can lead to a continued use of the system or technology [17]. Habit is one of the most influential factors *affecting* both Behavioral Intention and Use Behavior [19].
8. **Behavioral Intention (BI):** The degree to which a user has conscious plans to use or continue using the system or technology. It is a key indicator of future use behavior [17, 19].
9. **Use Behavior (UB):** The actual usage of the system or technology, typically measured through frequency or duration of use. If behavior tracking is not possible, it is often tracked through self-reported intention [17, 19].
10. **Trust (T):** Trust is one of the strongest predictors of acceptance in AI-integrated systems due to the uncertain nature of the algorithmic output [2]. Trust has been shown to significantly impact Behavioral Intention [12-14].

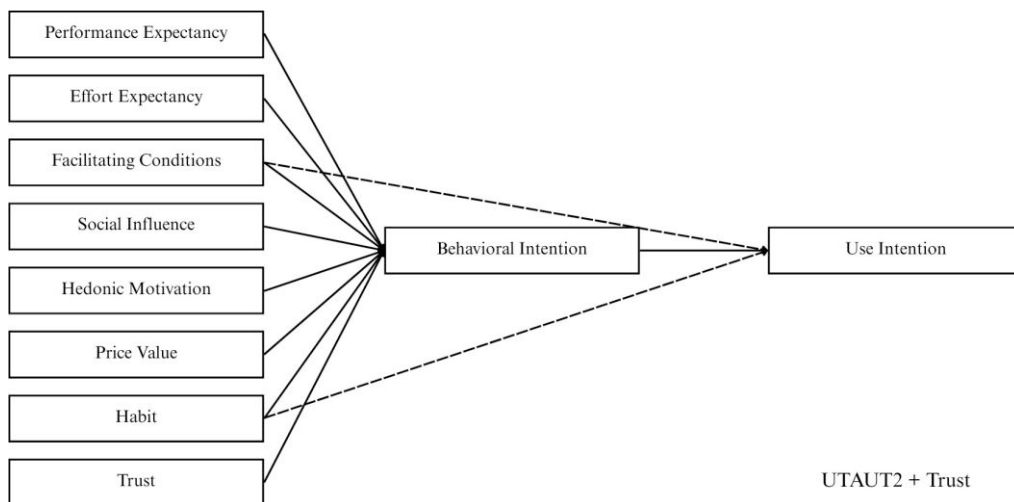


Fig. 2. Overview of the UTAUT2 model and their relationships with behavioral intention and use behavior.

For this study's purposes, the Facilitating Conditions and Price Value construct from the original UTAUT2 model was excluded. This decision is based on the nature of the application being a free mobile-based weather app designed for everyday use. Since the target users are smartphone users with adequate internet access and basic mobile app and technology literacy. As a result, Facilitating Conditions and Price Value were deemed unlikely to influence behavioral intention or use behavior in this context.

3.3. Data Analysis Technique

3.3.1. Quantitative Data Analysis Technique

To interpret the quantitative data from our questionnaire, our analysis will focus on descriptive statistics to measure tendencies, variability, and comparisons. Descriptive statistics that we used for this study:

1. **Mean:** Represents the average score from the data sample.
2. **Median:** The value separating the lower and upper half of the data sample. Data have to be sorted before finding the median. (n = the number of data)
3. **Mode:** The value that appears the most often in the data sample.
4. **Standard Deviation (SD):** Measures the dispersion or variability from the mean.
5. **p-Value:** The probability that the observed results occurred by chance. It's to test if differences between group means are statistically significant. If $p < 0.05$, it suggests a significant difference between groups.
6. **Cohen's d:** Used to measure the significance of differences between two group means. A Cohen's d of 0.2 is considered a small effect, 0.5 medium, and 0.8 large.
7. **T-Test:** Used to compare the means of two unrelated groups, which assesses whether the difference in scores between the two app types is statistically meaningful.
8. **Eta Squared:** Used to measure the significance of differences in the context of ANOVA. An Eta squared of 0.01 or less is considered a small effect, 0.06 a medium effect, and 0.14 or more a large effect.
9. **ANOVA:** Used to compare the means of more than two groups to test for significant differences.

The quantitative data analysis is processed through python for more accurate results, python libraries such as numpy and pandas are used.

3.3.2. Qualitative Data Analysis Technique

The qualitative data gathered from our questionnaire will analyzed through Thematic Analysis, which is used to identify and analyze themes within the data sample. Steps involved in Thematic Analysis:

1. Reading and re-reading data to become familiar with the data.
2. Make labels to identify important data of the data relevant to the research.
3. Gather labels into potential theme and gather all data relevant for each potential theme.
4. Check to see if the themes align with the entire dataset.
5. Refine each specific themes and the overall story of the dataset.
6. Final analysis, relat back to the research question.

3.4. Data Cleaning

Before conducting the main analysis, the raw survey data underwent a data cleaning process to ensure its quality and validity. The dataset was checked for duplicate submissions. Any duplicate records (repeated respondent identifiers) were identified and removed to ensure that each participant only contributed one valid response. Responses with suspicious or extreme values were also examined. Responses with uniform answers across all Likert-scale questions (straight-lining) or responses exhibiting contradictory answers were flagged and removed to maintain data integrity.

4. Experiment Results

4.1. Questionnaire

Our questionnaire consists of 3 separate sections: general, generic weather app review, and UI prototype weather app review. Below are the list of questions and options (for questions using multiple choice) from each section:

4.1.1. General

1. Gender: (Male/Female)
2. Current age: (<18, 18-26, 26-35, and >35)
3. Current employment: (student, employed, self-employed, and unemployed)
4. How often do you use weather applications? (couple times a day, once a day, couple times a week, rarely, and don't use at all)
5. What weather application are you currently using? (onboard weather application or third-party weather application)

4.1.2. Generic Weather App Scoring

Each questions are labeled correspondingly with their UTAUT2 construct and all questions below are answered using the five-point Likert scale.

1. PE 1 - I feel that the information provided by the application is quite useful
2. T - I can trust the information provided by the application
3. PE 2 - I feel that the application can help me in my daily activities
4. EE 1 - I feel that the application is easy to use
5. EE 2 - I can find the information I need easily
6. HT - I feel that I'm used to using the application to check the weather every day
7. HM 1 - I feel that the visual design (UI) is quite attractive and comfortable to look at
8. HM 2 - I feel that overall, the application is fun and interesting to use
9. SI - People around me (friends/family) encourage me to use this application
10. BI - I want to continue using the application

4.1.3. AI-based Weather App UI Prototype Scoring

Each questions are labeled correspondingly with their UTAUT2 construct and all questions below are answered using the five-point Likert scale, excepts for questions that are stated to be open-ended.

1. PE 1 - I feel that the information provided by the application is quite useful
2. T - I can trust the information provided by the application
3. PE 2 - I feel that the application can help me in my daily activities
4. EE 1 - I feel that the application is easy to use
5. EE 2 - I can find the information I need easily
6. HT - I feel that I can get used to using the application to check the weather every day
7. HM 1 - I feel that the visual design (UI) is quite attractive and comfortable to look at
8. HM 2 - I feel that overall, the application is fun and interesting to use
9. SI - If people around me (friends/family) use this application, I will also be interested in using it
10. BI 1 - I want to continue using the application
11. PE 3 - I feel that the AI features provided by the application will be useful for my daily activities
12. EE 3 - I feel that the AI features provided by the application will be easy to use
13. BI 2 - I might consider using this weather app over other weather apps if it were available now
14. What features and designs do you like the most/are most useful from the AI-based weather app? (open-ended)
15. What features and designs do you think can be improved or added for future weather app development? (open-ended)

The questionnaire is mostly aimed at a population of university students with ages ranging from 18 to 26 and aims to find out how the population feels about the weather app they are current using and the prototype weather app that we made, where the questions are based on the UTAUT2 constructs to better understand how it can correlate to user acceptance of weather applications as well as find out future directions for weather application development.

4.2. AI-Based Weather App UI Prototype

The weather app UI prototype that we developed focuses on AI features as a way to increase user acceptance of weather apps. Some of the features that we said are in the weather app can't be shown in our prototype since they are technical features related to the inner working that can't be shown as a UI. The features available in the app are:

1. AI personalization features in the form of chat-bots and suggestions to help choose outfits, places to go, and activities that are suitable for the current weather. In addition, it can also help provide context for weather data (e.g. why the weather is very hot today or what is meant by 20mm of rain). (This part is shown as a UI in the prototype)
2. Adaptive notifications that can provide weather information notifications at times when users usually open the application but will also provide notifications if there are drastic changes or weather warnings. (This is a feature that is said to be available in the weather app, but can't be directly shown)
3. Synchronization with the calendar application to provide weather context for every activity and event on the calendar (e.g. providing rain warnings and weather information on the day that has been planned in the calendar). (This part is shown as a UI in the prototype)
4. AI-based forecast system that provides local-specific weather information (weather forecasts according to the current specific location). Weather information will always be updated every hour and every day. (This is a feature that is said to be available in the weather app, but can't be directly shown)

Features that can't be shown in the prototype are communicated to the users through the questionnaire, so that participants know to score the prototype based on the features available. Figure 3 provides key illustrations of our UI weather app prototype. The full prototype used in this experiment is available at this URL: <https://www.figma.com/proto/5DUBKwpROxXh2zSddGXUqT/RM-SEM-4?node-id=0-1&t=ZZWWHQhiodeSlekt-1>

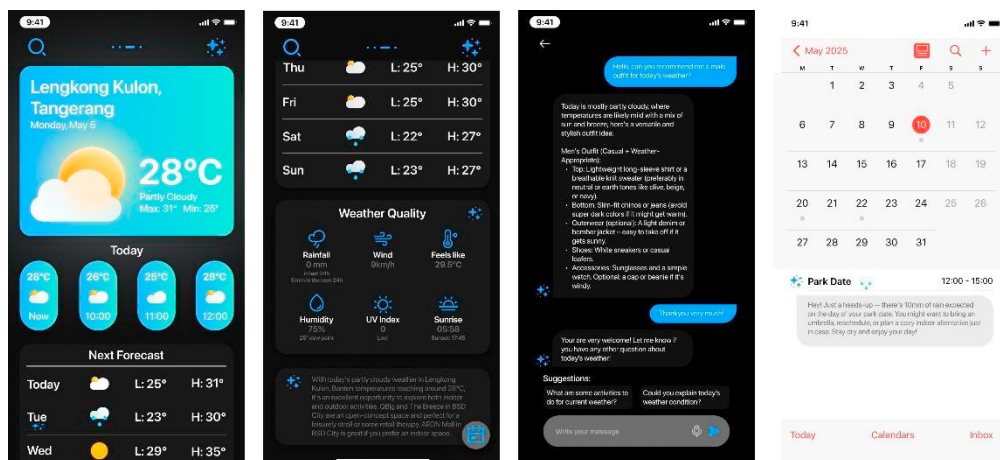


Fig. 3. Preview of the prototype which shows its' home screen, AI chat-bot, and calendar sync feature.

4.3. Results and Analysis

A total of 101 respondents filled out our questionnaire, which after data cleaning leaves 71 responses usable, 73.2% of which are male and 26.8% are female. 90.1% of the entire respondents were people age 18 - 25, while only 5.6% are age below 18, and only 4.2% are age 26 - 35. 87.3% of respondents are students, 5.6% are unemployed, while the other 7% are employed and self-employed. 39.4% of respondents claim to rarely use weather apps, 26.8% claim to use weather apps a couple times a week, 12.7% claim to use weather apps once a day, 14.1% claim to use weather apps a couple times a day, and the remaining 7% claim to not using weather apps. 91.5% of the respondents uses onboard weather apps, while only 8.5% uses third-party weather apps. Table 1 shows statistical results from the both scoring section of the questionnaire, comparing constructs such as Performance Expectancy (PE), Effort Expectancy

(EE), Trust (T), Habit (HT), and others. Figure 4 also visually highlights differences in user perceptions across constructs, showing users find higher satisfaction using an AI-integrated app.

Table 1: Questionnaire statistics overview - comparison of user responses to generic vs AI-based weather applications on key constructs.

Question Labels	Generic				AI-based			
	Mean	Median	Mode	SD.	Mean	Median	Mode	SD.
PE 1	4.01	4	4	0.74	4.22	4	4	0.66
T	3.76	4	4	0.89	4.05	4	4	0.73
PE 2	3.9	4	4	0.84	4.14	4	4	0.74
EE 1	4.35	4	4	0.62	4.39	4	5	0.65
EE 2	4.14	4	4	0.81	4.32	4	5	0.73
HT	3.71	4	4	1.07	4.10	4	4	0.76
HM 1	4.1	4	4	0.74	4.25	4	4	0.67
HM 2	3.90	4	4	0.81	4.18	4	4	0.76
SI	2.80	3	4	1.33	3.80	4	4	1.06
BI	3.69	4	4	0.90	3.90	4	4	0.87
PE 3					4.06	4	4	0.74
EE 3					4.28	4	5	0.78
BI 2					4.11	4	4	0.81

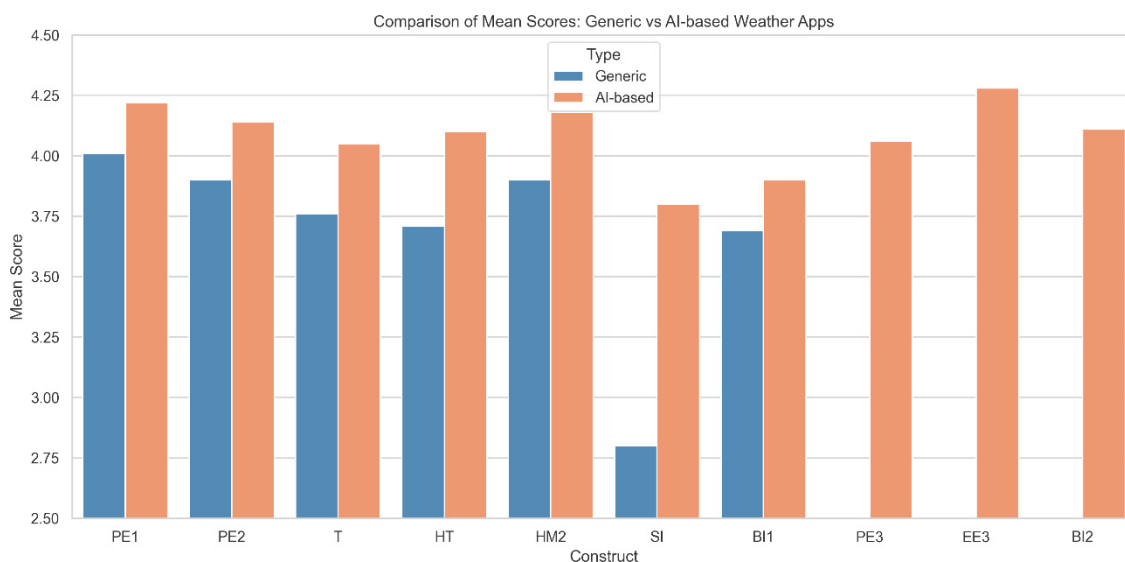


Fig. 4. Comparison of mean scores for UTAUT2 constructs between Generic and AI-based weather applications.

4.3.1. Quantitative Analysis

The comparative analysis between generic weather applications and AI-integrated weather applications show significant differences in multiple aspects of the UTAUT2 framework, as seen in Table 2. Users show a difference in their trust levels ($T: t = -2.238, p = 0.0267$), expressing more confidence in the reliability of information provided by the AI-integrated weather application. This suggests that respondents perceive the AI-integrated weather application as more reliable and credible in delivering accurate forecasts compared to generic weather apps. While the effect size is moderate, it demonstrates that AI capabilities such as personalized forecasting and adaptive interfaces can increase

users' confidence in the system. Furthermore, AI-integrated weather applications slightly outshine generic weather applications in hedonic motivation (HM2: $t = -2.223$, $p = 0.0277$), showing that users are enjoying the application experience. Users also show that they are a bit more likely to integrate an AI-integrated weather application into their daily routine, based on their habit-related responses (HT: $t = -2.646$, $p = 0.0091$), this shows that an AI-integrated weather application is effective in supporting daily planning. The most crucial difference is seen in social influence (SI: $t = -5.270$, $p < 0.0001$), which shows that users are significantly more influenced by their peers to adopt an AI-integrated weather application.

Table 2: Comprehensive statistical comparison between Generic vs AI weather apps and subgroup analysis. Bold values are referenced in the main findings paragraph.

Question Labels	Generic vs AI			Subgroup: Demographic			Subgroup: Usage Frequency		
	t-stat	p-value	Cohen's d	t-stat	p-value	Cohen's d	ANOVA	p-value	Eta Sq.
PE 1	-1.926	0.0559	-0.305	-1.073	0.2895	-0.263	1.206	0.3155	0.06
T	-2.238	0.0267	-0.354	-1.373	0.1767	-0.324	1.872	0.1242	0.091
PE 2	-1.901	0.0592	-0.301	-0.270	0.7883	-0.068	2.990	0.024	0.138
EE 1	-0.375	0.7081	-0.059	-2.255	0.0281	-0.489	1.237	0.3026	0.062
EE 2	-1.545	0.1243	-0.244	-0.171	0.8649	-0.044	0.577	0.6803	0.030
HT	-2.646	0.0091	-0.418	-2.272	0.0283	-0.554	0.423	0.7919	0.022
HM 1	-1.348	0.1795	-0.213	-1.697	0.0962	-0.393	1.213	0.3125	0.061
HM 2	-2.223	0.0277	-0.351	-1.343	0.1862	-0.321	0.407	0.8033	0.021
SI	-5.270	0.0000	-0.833	-1.601	0.1166	-0.383	1.407	0.2400	0.070
BI 1	-1.503	0.1349	-0.238	-2.010	0.0498	-0.427	0.735	0.5713	0.038
PE 3	—	—	—	-3.121	0.0032	-0.746	0.817	0.5183	0.042
EE 3	—	—	—	-1.278	0.2073	-0.289	1.423	0.2347	0.071
BI 2	—	—	—	-1.086	0.2831	-0.257	0.615	0.6534	0.032

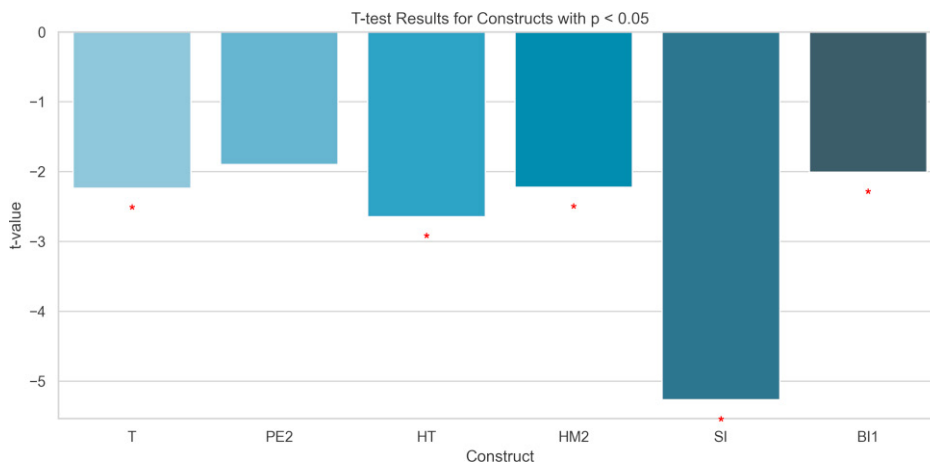


Fig. 5. Significant differences in UTAUT2 constructs ($p < 0.05$) between generic and AI-based apps.

As seen in Figure 5, Trust (T), Habit (HT), Hedonic Motivation (HM2), and Social Influence (SI) all have significant differences in favor of AI-based applications. This figure summarizes the constructs that had p-values below 0.05, showing the significance of these differences.

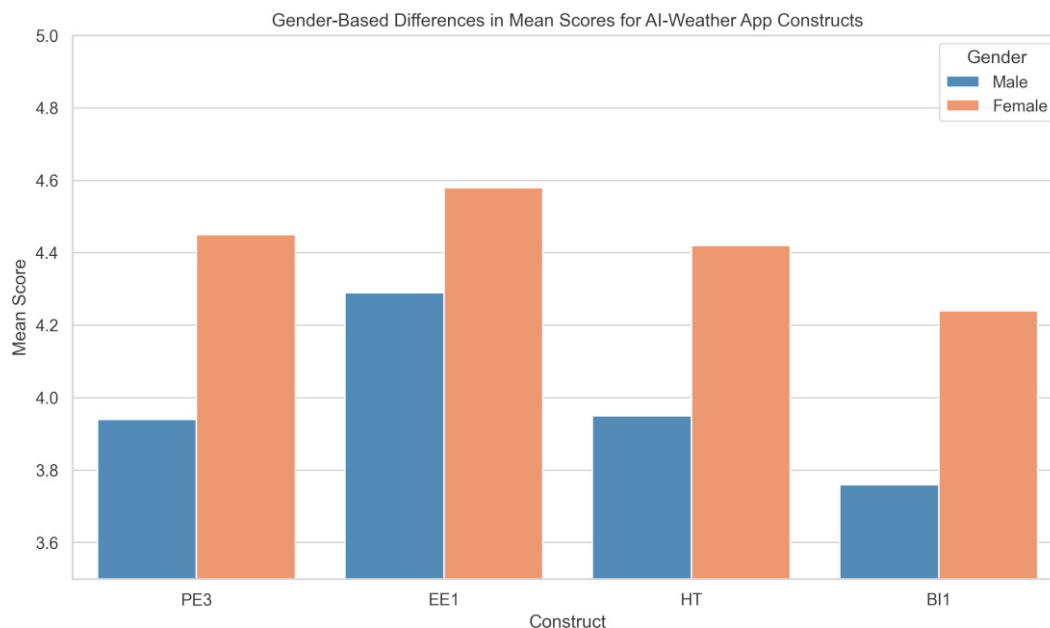


Fig. 6. Gender-based differences in perception of AI-based weather app features.

For user perception towards AI-integrated weather applications, PE3, is seen having a significant difference between genders as seen in Figure 6. Female participants agree more that AI-integrated features are beneficial for daily activities (PE3: $t = -3.121$, $p = 0.0032$). There is, however, no significant difference between EE3 and BI2, showing that both gender demographics have similar opinions on the matter. Notably, habit (HT), and behavioral intention (BI1) also exhibit statistically significant gender differences, with female participants showing higher agreement on the habitual use of the app (HT: $t = -2.272$, $d = 0.0283$) and a stronger intention to continue using it (BI1: $t = 2.010$, $p = -0.0498$). This suggests that female users are more likely to integrate the AI-based weather application into their daily routine and continue engaging with it over time. It is also worth noting from ease expectancy (EE 1), female participants have an easier time with using the app (EE1: $t = -2.255$, $p = -0.0281$). However, as the sample is predominantly male (73.3%), these gender-based findings should be interpreted with caution due to the unequal group sizes. Additional data from more balanced gender distributions may be needed to confirm these effects.

When segmented by usage frequency, the analysis shows that only PE2 demonstrates a statistically significant difference between user groups (PE2: ANOVA = 2.990, $p = 0.0234$, $\eta^2 = 0.11$). This suggests that how often users check weather applications affects their perception of how helpful the AI-integrated weather app is for supporting daily activities. Specifically, users who check the weather less frequently tend to perceive the AI features as slightly more beneficial for their daily tasks, compared to frequent users who may have higher expectations or rely on other sources. For all other constructs, including ease of use (EE1, EE2, EE3), habit (HT), hedonic motivation (HM1, HM2), trust (T), social influence (SI), and behavioral intention (BI1, BI2), no significant differences were found across usage frequency groups. Age group and employment comparisons are inconclusive due to incomplete data, as these demographics are either largely dominated by certain groups, or have certain demographics that did not participate, suggesting a need for more diverse demographic data in future studies. These results suggest that AI enhancements to a weather application positively impact key determinants of technology acceptance when measured under the UTAUT2 framework, increasing the user base's willingness to integrate weather applications enhanced by AI.

4.3.2. Qualitative Analysis

Based on the qualitative responses gathered, several recurring themes are seen regarding features that are perceived as useful as well as features that have areas for improvement. Several users indirectly referenced trust as an important factor. Users show appreciation for real-time accurate forecasts that update on an hourly basis, showing interest in detailed and timely weather updates. Another appreciated features are the AI-powered chatbot, which offers personalized advice such as clothing suggestions or aid in planning daily activities based on the current weather. Thus showing that users have a strong appreciation for personalization and customization, valuing a weather application that adapts the information provided to the user base's needs and priorities. The AI chatbot's ability to provide personalized suggestions also contributed to users' perception of reliability. This reinforces the quantitative findings, emphasizing that trust is not only measured in data but also perceived experientially through consistent, personalized, and transparent features. Calendar integration is also a positive since users are enabled to plan their activities much more easily.

As for the field of improvements, users seek increased visual clarity, further interactivity, greater accessibility, and more proactivity. Users express desire to see a more colorful interface, requesting an animated or even an adaptive background, one that reflects the current weather or an extra feature such as a light and dark mode to fit the users preference. Enhanced interactivity is also needed in order for users to have a much more enjoyable experience, users have an interest in expanding the scope of information provided. Several users suggest the addition of air quality indicators, cloud movement analysis, and explanations behind weather patterns. Some users stated the need for tutorials or simplified UI to help less tech-savvy users. Users also emphasize their desire for a more proactive system such as custom notifications, this could increase user engagement and trust in the application's predictions. These insights highlight that users are receptive to and even enthusiastic for AI-integrated weather applications, which is capable of delivering a customizable, personalized, concise, and engaging. Users also emphasized the need for expanded metrics such as air quality indicators and cloud movement analysis, which would enhance the app's utility for planning outdoor activities. Additionally, requests for hyper-local predictions (e.g., street-level forecasts) and explanatory AI features (e.g., summaries of humidity trends) further highlight the importance of precision and transparency in fostering trust. These results suggest that AI enhancements to weather applications positively impact key determinants of technology acceptance under the UTAUT2 framework. Notably, trust emerged as a significant factor, with users reporting greater confidence in AI-powered forecasts, which contributes to their overall intention to adopt and rely on the application. This trust is built through personalized interaction, timely updates, and the perceived intelligence of the system, suggesting that AI plays a critical role not only in functionality but also in building user confidence, which is essential for long-term engagement.experience.

5. Discussion and Conclusion

Our findings reveal that AI integration meaningfully improves key acceptance indicators such as Performance Expectancy, Trust, Hedonic Motivation, and Habit compared to generic weather applications. Users demonstrated higher satisfaction, a stronger intent to use the app, and greater appreciation for personalized and contextual features. The statistical significance of these findings confirms that AI-based enhancements can address concerns of engagement, usability, and trust in weather applications. While the study provides substantial evidence that AI features can elevate user experience and behavioral intention, it is not without limitations. The sample was predominantly composed of university students within a narrow age range. The distributed questionnaire may be filled out by the same respondent more than once, which leads to the need for data cleaning. Moreover, the prototype was evaluated based on perceived features rather than real-time usage, which may not fully capture behavioral patterns or system reliability.

Future research should aim to expand the demographic diversity of participants to include older adults, different social/economic groups, and users in varying geographic regions. Additionally, implementing a functional version of the AI-enhanced weather application for longitudinal studies would allow deeper insights into user behavior and acceptance. Future studies could also investigate the impact of AI transparency (clearer explanations) and feedback loops affect users' trust and satisfaction over time. Integrating additional constructs such as perceived risk or privacy concern may further refine the UTAUT2-Trust framework for evaluating AI-integrated services.

AI Usage Declaration

We declare that we used ChatGPT (OpenAI) to assist with idea generation and data extraction during the development of this manuscript. All generated content was critically reviewed and refined by the authors to ensure accuracy, relevance, and appropriate academic tone. The research design, methodology, analysis, and core contributions remain entirely the original work of the authors.

Author Contribution Statement

M. L. Nahtan led the study's framework development, questionnaire development, questionnaire distribution, data processing algorithm, and data analysis. S. Lukmana led the conceptualization, review of the literature, and analysis of the study. A. D. Novika, S.Kom., M.Kom. and A. C. Sari, S.Kom., M.T.I. oversaw the development of the study as well as supervision. I. L. P. S. Rusli concluded the study.

Data Availability Statement

The data supporting the findings of this study were collected through a self-administered questionnaire distributed online to voluntary participants, and are publicly available on GitHub at: <https://github.com/marpinnn/Driving-User-Acceptance-of-AI-Enhanced-Weather-Applications->. The UTAUT2 framework that was adapted and used for analysis and data processing was adapted from existing literature by Venkatesh et al. (2012).

References

- [1] Bryant MJ, Wilde SJ, Smart WJ. Taking the weather with you: user acceptance, trust and value of weather apps on smartphones. *IJSHC*. 2017;2(3/4):247.
- [2] Vaughn C, Sherman-Morris K, Brown M. A Change in the Weather: Understanding Public Usage of Weather Apps. *J Operational Meteor*. 2023 Dec 29;140–60.
- [3] Mohd Rahim NI, A. Iahad N, Yusof AF, A. Al-Sharafi M. AI-Based Chatbots Adoption Model for Higher-Education Institutions: A Hybrid PLS-SEM-Neural Network Modelling Approach. *Sustainability*. 2022 Oct 6;14(19):12726.
- [4] Hasan MR, Chowdhury NI, Rahman MH, Syed MAB, Ryu J. Understanding AI Chatbot adoption in education: PLS-SEM analysis of user behavior factors. *Computers in Human Behavior: Artificial Humans*. 2024 Aug;2(2):100098.
- [5] Yildiz Durak H, Onan A. Predicting the use of chatbot systems in education: a comparative approach using PLS-SEM and machine learning algorithms. *Curr Psychol*. 2024 Jul;43(28):23656–74.
- [6] Liao QV, Gruen D, Miller S. Questioning the AI: Informing Design Practices for Explainable AI User Experiences. In: *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* [Internet]. Honolulu HI USA: ACM; 2020 [cited 2025 May 14]. p. 1–15. Available from: <https://dl.acm.org/doi/10.1145/3313831.3376590>
- [7] Bi K, Xie L, Zhang H, Chen X, Gu X, Tian Q. Pangu-Weather: A 3D High-Resolution Model for Fast and Accurate Global Weather Forecast [Internet]. *arXiv*; 2022 [cited 2025 May 14]. Available from: <https://arxiv.org/abs/2211.02556>
- [8] Chen K, Bai L, Ling F, Ye P, Chen T, Luo JJ, et al. Towards an end-to-end artificial intelligence driven global weather forecasting system [Internet]. *arXiv*; 2023 [cited 2025 May 14]. Available from: <https://arxiv.org/abs/2312.12462>
- [9] Rackow T, Koldunov N, Lessig C, Sandu I, Alexe M, Chantry M, et al. Robustness of AI-based weather forecasts in a changing climate [Internet]. *arXiv*; 2024 [cited 2025 May 14]. Available from: <https://arxiv.org/abs/2409.18529>
- [10] Weyn JA, Durran DR, Caruana R. Improving Data-Driven Global Weather Prediction Using Deep Convolutional Neural Networks on a Cubed Sphere. *J Adv Model Earth Syst*. 2020 Sep;12(9):e2020MS002109.
- [11] Radford JT, Ebert-Uphoff I, Stewart JQ. A Comparison of AI Weather Prediction and Numerical Weather Prediction Models for 1–7-Day Precipitation Forecasts. *Weather and Forecasting*. 2025 Apr;40(4):561–75.
- [12] Bach TA, Khan A, Hallock H, Beltrao G, Sousa S. A Systematic Literature Review of User Trust in AI-Enabled Systems: An HCI Perspective. *International Journal of Human–Computer Interaction*. 2024 Mar 3;40(5):1251–66.
- [13] Vimalkumar M, Sharma SK, Singh JB, Dwivedi YK. ‘Okay google, what about my privacy?’: User’s privacy perceptions and acceptance of voice based digital assistants. *Computers in Human Behavior*. 2021 Jul;120:106763.
- [14] Choung H, David P, Ross A. Trust in AI and Its Role in the Acceptance of AI Technologies. 2022 [cited 2025 May 14]; Available from: <https://arxiv.org/abs/2203.12687>
- [15] Vaughn C, Sherman-Morris K, Brown M, Gutter B. That’s not what my app says: Perceptions of accuracy, consistency, and trust in weather apps. *Meteorological Applications*. 2024 May;31(3):e2205.
- [16] Di Fant V, Del Pozo M, Gulikers J, Paparrizos S. Conceptualizations and implementation of user engagement in Weather and climate services: A climate services providers’ perspective. *Heliyon*. 2024 Jan;10(1):e22940.
- [17] Zhang ZF, Haji Darmi R, Yap NT, Nimehchisalem V. Extended Unified Theory of Acceptance and Use of Technology in Mobile Learning: A

- Systematic Review. *IJARPED*. 2022 Sep 17;11(3):846-865.
- [18] Prihanto IG, Gunawan H, Riyanto B, Prasetyo W, Sumaedi S, Rakhmawati T, et al. A technology acceptance model of satellite-based hydrometeorological hazards early warning system in Indonesia: an-extended technology acceptance model. *Cogent Business & Management*. 2024 Dec 31;11(1):2374880.
 - [19] Tamilmani K, Rana NP, Dwivedi YK. Consumer Acceptance and Use of Information Technology: A Meta-Analytic Evaluation of UTAUT2. *Inf Syst Front*. 2021 Aug;23(4):987–1005.
 - [20] Mlekus L, Bentler D, Paruzel A, Kato-Beiderwieden AL, Maier GW. How to raise technology acceptance: user experience characteristics as technology-inherent determinants. *Gr Interakt Org*. 2020 Sep;51(3):273–83.
 - [21] McGovern A, Demuth J, Bostrom A, Wirz CD, Tissot PE, Cains MG, et al. The value of convergence research for developing trustworthy AI for weather, climate, and ocean hazards. *npj Nat Hazards*. 2024 Jul 1;1(1):13.
 - [22] Bakhar M, Fajri TI, Sumerli A CH, Ahmad N, Nuryanto UW. User Acceptance Analysis of My Pertamina Application Using UTAUT Model. *jsisfotek [Internet]*. 2023 Jul. 10 [cited 2025 May 14];5(2):182-6. Available from: <https://jsisfotek.org/index.php/JSisfotek/article/view/254>